Preprint Notice

Artificial Intelligence of Things–Enabled Sleep Monitoring Using Wearable Inertial Sensors and a Transformer–CNN Model

Hamza Ali Imran, Ataul Aziz Ikram, Saad Wazir

***Abstract*—To enable Artificial Intelligence of Things (AIoT) for scalable sleep monitoring, we present Neendh-T, a compact Transformer–CNN model for sleep–wake detection from wrist-worn actigraphy. Using the MESA Sleep dataset and the official Night and Night&Day benchmark protocols, Neendh-T achieves 83.25% accuracy and 85.98% F1 on Task 1, and 87.58% accuracy and 82.28% F1 on Task 2. Compared with the non-rescored MESA baselines, Neendh-T improves sleep F1 and sensitivity relative to LSTM-100 while maintaining competitive accuracy. For AIoT deployment, the FP32 TFLite model reaches 1.91 ms on a Xiaomi 13T, 3.07 ms on a Raspberry Pi 4 Model B, and 0.396 ms on an Intel i5–13420H CPU, corresponding to about 6–7× speed-up over our LSTM-100 deployment of benchmark model on mobile and edge platforms. These results show that Neendh-T balances classification performance and low-latency on-device inference without cloud support**

***Index Terms*—AI sensors, Artificial Intelligence of Things (AIoT), Sleep Monitoring, Actigraphy, Wearable Sensors, Inertial Measurement Unit (IMU), Sleep–Wake Detection, MESA Sleep Dataset.**

I. Introduction

Sleep is critical to both physical and mental health, but reliable large-scale sleep quality evaluation outside of laboratory conditions has been difficult to achieve. Polysomnography (PSG), referred to as the clinical gold standard, provides high-resolution physiological data but is highly expensive and unsustainable for long-term population-level monitoring [1]. A low-cost, non-invasive, and widely adopted alternative is to use wearable Inertial Measurement Units (IMUs) [2], [3]. However, today's commercial actigraphy is mostly based on proprietary rule-based heuristics with elementary motion thresholds and smoothing filters that do not work well across widely varying populations, complex sleep–wake patterns, or low-movement sleepers [4]. This restrict calls for open, data-driven, and generalisable methods to sleep–wake modelling. The fusion of embedded sensing with lightweight machine learning to facilitate Artificial Intelligence (AI) was coined as Artificial Intelligence of Things, where inference running on-device in real-time is now feasible [5]–[7]. In this regard, wrist-band IMUs can enable continuous 24-hour sleep–wake estimation with enhanced privacy and reduced transmission overhead, providing a scalable pathway for personalized health monitoring as well as the conduct of large-scale epidemiological studies. Despite these advances, three gaps remain important for embedded sleep monitoring. First, many sleep–wake models are evaluated primarily as offline classifiers, with limited attention to on-device latency and deployability. Second, benchmark comparisons can be affected by differences in preprocessing, windowing, and post-processing rules. Third, recurrent baselines such as LSTMs provide strong performance but rely on sequential computation, which can limit low-latency inference on heterogeneous AIoT platforms. This motivates a compact model that preserves temporal context, captures local motion bursts, and remains efficient under embedded inference settings. The MESA Sleep dataset [8], with a large and diverse cohort of concordant PSG and actigraphy recordings, is particularly well-suited for AIoT-targeted modelling. Following Palotti et al. [9], we use the Night and Night&Day benchmark tasks for evaluation. We propose *Neendh-T*, a lightweight Transformer–CNN hybrid model for wearable IMU sleep monitoring. Its combination of long-range self-attention and multi-scale convolution enables accurate, energy-efficient on-device sleep–wake detection suitable for AIoT deployments. The key contributions of this work are:

- A compact Transformer–CNN model, Neendh-T, for sleep–wake detection from wrist-worn actigraphy, achieving **83.25%** accuracy and **85.98%** F1 on MESA Task 1, and **87.58%** accuracy and **82.28%** F1 on Task 2.
- Evaluation under the MESA Task 1 and Task 2 protocols with explicit window-to-epoch prediction mapping and no rule-based post-processing, including no Webster rescoring.
- FP32 TFLite deployment benchmarking across smartphone, Raspberry Pi, and desktop CPU platforms, achieving **1.91 ms**, **3.07 ms**, and **0.396 ms** mean latency, respectively, with about **6–7**× speed-up over our LSTM-100 deployment reference on mobile and edge platforms.

Corresponding author: H. A. Imran (e-mail: engrhamzaaliimran@gmail.com). Hamza Ali Imran, Ataul Aziz Ikram, and Saad Wazir are with the School of Creative Technologies, University of Bolton, Islamabad Campus, Pakistan. Saad Wazir is also with the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.

II. Related Work

The MESA benchmark by Palotti et al. established standardized Night and Night&Day sleep–wake tasks and compared traditional actigraphy rules, classical machine-learning models, and deep-learning baselines. Traditional rules are computationally simple but often favor sleep sensitivity over wake specificity, while CNN and LSTM models learn temporal representations directly from actigraphy windows. However, the benchmark primarily evaluates classification performance rather than AIoT-oriented deployment latency. Our work retains the benchmark protocol but adds a compact Transformer–CNN model, explicit post-processing assumptions, and embedded inference measurements.

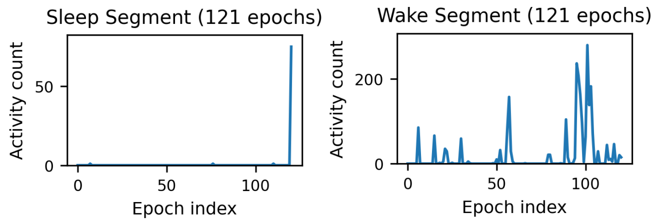

Fig. 1: Representative raw activity-count sequences from the MESA actigraphy dataset. The left panel shows a 121-epoch ($\approx$ 1 h) sleep segment (mesaid = 2) characterised by long stretches of near-zero activity with only sparse micro-movements. The right panel shows a 121-epoch wake segment (mesaid = 1) exhibiting high-amplitude, high-variance fluctuations typical of daytime motor behaviour.

III. Methodology

A. Dataset and Pre-processing

We use the Multi-Ethnic Study of Atherosclerosis (MESA) Sleep cohort which includes 30 s resolution annotations that synchronize wrist-worn actigraphy and polysomnography (PSG) [8], [10]. The actigraph is equipped with a low-power tri-axial accelerometer and does not retain raw data, instead performing all high-frequency samples into a single 30,s *activity count*, providing a one-dimensional (1D) motion signal invariant to orientation appropriate for long-term wearable sensing. Several significant behavioural signatures were identified using actigraphy (Fig. 1): sleep consists of long near-zero activity intervals with sparse micro-movements, while wake exhibits high-amplitude, high-variance fluctuations driven by voluntary motion. These differences reflect the wrist's stationary nature during sleep versus dynamic accelerations during daytime activity. Such temporal structure motivates temporal deep-learning models—such as the hybrid Transformer–CNN design of Neendh-T to capture both long-range behaviour and short-duration motion bursts for sleep–wake discrimination.

All MESA recordings are temporally aligned, converted to epoch-wise samples, and merged with annotations. Following Palotti et al. [9], we define two tasks: **Task 1** (Night) uses only PSG-window epochs, with sleep defined as any stage > 0; **Task 2** (Night&Day) adds up to 960 epochs ($\sim$8 h) of daytime context. For non-PSG intervals, sleep is inferred from REST/REST-S labels, while PSG supervises nighttime data. After filtering incomplete recordings, the cohort includes **1832 subjects**, split subject-wise into **1465** (80%) for training and **367** (20%) for testing, using the same split for both tasks. This large, clinically validated dataset provides a strong basis for developing lightweight, AIoT-ready sleep–wake models under both PSG-supervised and context-augmented settings. For model input generation, each sample is formed as a 121-epoch actigraphy window, corresponding to approximately 60.5 min of 30 s activity-count epochs. The prediction target is the center epoch of the window. During evaluation, sliding windows are generated over each test recording, and each eligible center epoch receives one prediction from its corresponding centered window. Boundary epochs that cannot form a full centered window are excluded consistently from both prediction and metric computation. Therefore, overlapping windows are not reconciled using voting or averaging; each evaluated epoch has exactly one model output.

B. Neendh-T: Proposed Architecture

Neendh-T is shown in (Fig. 2). Each input sample is a fixed-length actigraphy window

$$X \in \mathbb{R}^{121\times 1}$$

$$H_0 = \text{Dense}(64)(X) + PE \tag{1}$$

Temporal dependencies are captured using two stacked pre-normalized Transformer encoder blocks with four attention heads ($d_k = 16$). For intermediate representation H:

$$H' = H + \text{Dropout}(\text{MHA}(\text{LN}(H))), \tag{2}$$

$$R = H' + \text{Dropout}(\text{Dense}(256, \text{ReLU})(\text{LN}(H')) \to \text{Dense}(64)) \tag{3}$$

After two blocks, $R_0 \in \mathbb{R}^{121\times 64}$. Short-term actigraphy dynamics (e.g., micro-movements, brief restlessness) are modeled using three sequential multi-scale temporal convolution modules. For module i with input R_{i-1}:

$$B_{k,i} : \text{Conv1D}(16, 1, \text{ReLU}) \to \text{Conv1D}(16, k, \text{ReLU}), \quad k \in \{3, 5, 7\}. \tag{4}$$

Aggregated responses are projected and combined residually:

$$R_i = \text{Conv1D}(96, 1)\left(\sum_k B_{k,i}\right) + \text{Conv1D}(96, 1)(R_{i-1}), \tag{5}$$

yielding $R_3 \in \mathbb{R}^{121\times 96}$. A final pointwise convolution refines the representation:

$$H_f = \text{Conv1D}(128, 1, \text{ReLU})(R_3), \tag{6}$$

followed by global average pooling $z = \text{GAP}(H_f) \in \mathbb{R}^{128}$. Binary sleep–wake probabilities are produced via

$$\hat{y} = \text{Softmax}(Wz + b). \tag{7}$$

Here, `Dense(64)` projects each scalar activity-count epoch into a 64-dimensional embedding, and `PE` adds sinusoidal positional information across the 121-epoch sequence. `LN`, `MHA`, and `GAP` denote layer normalization, multi-head attention, and global average pooling. $\mathbf{H}_0$ and $\mathbf{H}$ denote Transformer-stage feature maps, $\mathbf{R}_i$ denotes the residual CNN feature maps, and $\mathbf{H}_f$ is the final pointwise-projected representation before pooling. The final binary class prediction is obtained using an argmax operation over the two softmax probabilities. No probability-threshold tuning, smoothing, majority voting, Webster rescoring, or other post-processing is applied.

IV. Results

Neendh-T was evaluated under the two MESA benchmark tasks proposed by Palotti et al., **Task 1 (Night)** and **Task 2 (Night&Day)**. All epoch-level metrics were computed using the subject-wise split defined in the benchmark. In accordance with scikit-learn's binary classification convention, the reported F1-score corresponds to the *positive class* (Sleep), matching the evaluation methodology of the original benchmark.

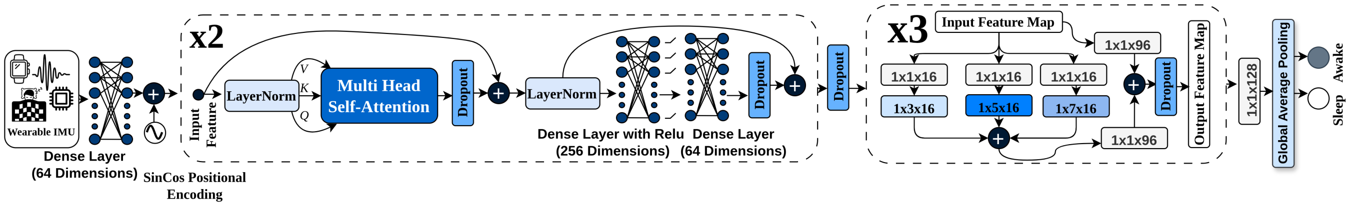

Fig. 2: Neendh-T architecture for sleep–wake detection. A 121×1 actigraphy sequence is processed using self-attention and multi-scale temporal convolutions, followed by pointwise convolution, global average pooling, and classification.

TABLE I: {Comparison of Neendh-T with the strongest non-rescored MESA benchmark baselines.

Benchmark values taken from Palotti *et al.*, npj Digital Medicine (2019).

Method	Accuracy	F1 (Sleep)
Task 1 (Night)		
Device Algorithm (Oakley θ=40)	76.2%	81.3%
Extra Trees (best ML)	81.8%	84.3%
LSTM-100 (best DL)	83.1%	85.5%
Neendh-T (ours)	**83.25%**	**85.98%**
Task 2 (Night&Day)		
Device Algorithm	76.6%	71.2%
Extra Trees (best ML)	86.7%	77.3%
LSTM-100 (best DL)	88.2%	80.8%
Neendh-T (ours)	**87.58%**	**82.28%**

TABLE II: Epoch-level performance of Neendh-T on MESA Task 1 and Task 2.

Metric	Task 1	Task 2
Accuracy	83.25%	87.58%
Precision (Sleep)	80.81%	71.93%
Sensitivity (Sleep)	91.85%	96.10%
Specificity (Wake)	72.35%	83.93%
F1-score (Sleep)	85.98%	82.28%
ROC-AUC	0.8938	0.9560
PR-AUC (AP)	0.8851	0.8699

A. Overall Performance

Table IV shows that Neendh-T achieves performance at or above the strongest baselines reported in the benchmark study [9]. For **Task 1**, Neendh-T matches the best deep-learning model (LSTM-100) in accuracy and surpasses it in F1-score. For **Task 2**, Neendh-T maintains competitive accuracy while achieving the *highest* sensitivity and F1 among all methods evaluated, including traditional, ML, and DL algorithms. The detailed performance extracted from our evaluation logs is shown in Table II.

B. Confusion Matrix Interpretation

Fig. 3 presents the confusion matrices for both tasks.

1) Task 1 (Night): Neendh-T correctly identifies **91.85% of sleep epochs** (242,165 out of 263,649) and **72.35% of wake epochs** (150,432 out of 207,934). This behavior closely mirrors the benchmark's strongest DL models, which also demonstrate high sleep sensitivity but more modest wake specificity during nocturnal periods. Compared with the device algorithm (50.1% specificity), Neendh-T provides substantially improved wake detection—a known weakness of traditional actigraphy methods.

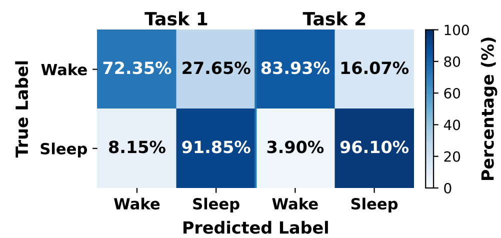

Fig. 3: Combined confusion matrices for *Neendh-T* on both tasks. The left panel corresponds to **Task 1 (Night)**, and the right panel corresponds to **Task 2 (Night&Day)**. Each cell reports the percentage of predictions, with the absolute sample count shown in parentheses.

2) Task 2 (Night&Day): The 24-hour Task 2 is more heterogeneous, yet Neendh-T achieves a **96.10% sleep sensitivity**, the highest among all models, including LSTM-100 (86.4%). Wake specificity also remains strong at **83.93%**. These results indicate that Neendh-T successfully leverages daytime movement patterns to maintain balanced discrimination, outperforming both the device algorithm and ML baselines in F1-score while closely matching the accuracy of the strongest DL benchmark model.

C. Comparison with the MESA Benchmark

The benchmark study [9] reports both non-rescored and Webster-rescored results. To avoid mixing learned-model performance with rule-based temporal post-processing, Neendh-T is compared only with the non-rescored baselines. Under this setting, traditional algorithms achieve 70–78% accuracy for Task 1 and 70–83% for Task 2, while the device algorithm attains 76.2% and 76.6%, respectively. Neendh-T improves accuracy over the device algorithm on both tasks and improves F1-score relative to the non-rescored Extra Trees and LSTM-100 baselines, while maintaining competitive Task 2 accuracy. These results indicate that a compact Transformer–CNN model can provide strong sleep–wake discrimination without applying Webster rescoring or any other post-processing.

D. Ablation Study

We evaluated CNN-only, Transformer-only, and the full Neendh-T hybrid model. The CNN-only variant removes the encoder blocks, retaining the multi-scale convolutions, while the Transformer-only model removes the CNN modules and consists solely of stacked self-attention encoders with positional encoding. We further implemented a 1D MobileNetV3-Small adaptation [11], replacing 2D spatial convolutions with

TABLE III: Ablation study on MESA Task 1 (Night).

Model	Accuracy	F1 (Sleep)	Specificity (Wake)
CNN-only	78.75%	83.77%	54.19%
Transformer-only	82.90%	85.37%	74.86%
MobileNetV3 (1D)	80.82%	83.30%	74.84%
Neendh-T (Hybrid)	83.25%	85.98%	72.35%

TABLE IV: Ablation study on MESA Task 2 (Night&Day).

Model	Accuracy	F1 (Sleep)	Sensitivity (Sleep)
CNN-only	84.70%	79.20%	97.07%
Transformer-only	89.23%	83.89%	93.50%
MobileNetV3 (1D)	88.78%	82.51%	88.20%
Neendh-T (Hybrid)	87.58%	82.28%	96.10%

TABLE V: FP32 inference performance and parameter-count comparison of *Neendh-T* and the LSTM-100 reference on mobile, edge, and fog platforms. Inference uses the XNNPACK CPU delegate with 4 threads. Speed-up is computed as $\text{Latency}_{\text{LSTM}}/\text{Latency}_{\text{Neendh-T}}$.

Platform	Model	Mean (ms)	Inf/s	Speed-up (×)
Xiaomi 13T	Neendh-T	1.91	520	7.26×
Xiaomi 13T	LSTM-100	13.86	72	
RPi 4 Model B (CPU)	Neendh-T	3.07	326	6.56×
RPi 4 Model B (CPU)	LSTM-100	20.13	50	
i5–13420H (Linux)	Neendh-T	0.396	2528	2.52×
i5–13420H (Linux)	LSTM-100	0.997	1003	

temporal operations while preserving inverted residual and squeeze-and-excitation modules. The evaluated ablation models contain 66,658 parameters for CNN-only, 67,202 for Transformer-only, 166,626 for Neendh-T, and 1,423,462 for MobileNetV3 (1D). Thus, Transformer-only is smaller than Neendh-T and performs competitively, particularly on Task 2. However, Neendh-T is retained because it provides the preferred sleep-sensitive operating point: it achieves the highest Task 1 F1-score and the highest Task 2 sleep sensitivity. This suggests that self-attention captures long-range temporal context, while the multi-scale CNN stage further refines local motion patterns. Therefore, Neendh-T is not selected because it dominates every metric, but because it offers a balanced accuracy–sensitivity trade-off for continuous AIoT sleep monitoring.

E. On-Device and Edge Inference Performance

{To evaluate AIoT deployability, we benchmarked the FP32 TFLite implementation of *Neendh-T* across mobile, edge, and fog platforms using the XNNPACK CPU delegate with four threads. The LSTM-100 reference was implemented using the same input representation and inference settings, so the latency comparison should be interpreted as an implementation-controlled deployment reference. On a Xiaomi 13T smartphone, *Neendh-T* achieves a mean latency of **1.91 ms** (median 1.53 ms), corresponding to **520 inferences/s**, compared with **13.86 ms** (≈72 inferences/s) for LSTM-100. On a Raspberry Pi 4 Model B, *Neendh-T* achieves **3.07 ms** (median 3.00 ms), compared with **20.13 ms** for LSTM-100. On an Intel Core i5–13420H workstation, *Neendh-T* achieves **0.396 ms**, compared with **0.997 ms** for LSTM-100. These results show that parameter count alone does not determine deployment efficiency: although LSTM-100 has fewer parameters than Neendh-T, its recurrent sequential computation leads to higher latency. Overall, the parallelizable Transformer–CNN design enables low-latency inference across heterogeneous AIoT platforms without reliance on cloud resources. Table V summarizes the results.

V. Conclusion and Future Work

In this paper, we developed Neendh-T for sleep–wake detection using wearable IMU actigraphy. The model performs strongly on MESA Task 1 and Task 2 and runs in real time on smartphones and edge CPUs, indicating that continuous sleep monitoring is feasible without cloud support or specialised hardware. While limited to binary sleep–wake classification, this work provides a basis for future extensions, including multistage sleep prediction, on-device adaptation, self-supervised learning from unlabeled actigraphy, and explainability methods such as attention-based relevance maps.

References

[1] D. Vitazkova, H. Kosnacova, D. Turonova, E. Foltan, M. Jagelka, M. Berki, M. Micjan, O. Kokavec, F. Gerhat, and E. Vavrinsky, "Transforming sleep monitoring: review of wearable and remote devices advancing home polysomnography and their role in predicting neurological disorders," *Biosensors*, vol. 15, no. 2, p. 117, 2025.

[2] J. Matos, B. Ramos, J. Fernandes, C. Hansen, W. Maetzler, N. Vila-Chã, and L. F. Maia, "Wearable sensors for sleep monitoring in free-living environments: a scoping review on parkinson's disease," *Biosensors*, vol. 15, no. 4, p. 212, 2025.

[3] A. Logacjov, K. Bach, and P. J. Mork, "Long-term self-supervised learning for accelerometer-based sleep–wake recognition," *Engineering Applications of Artificial Intelligence*, vol. 141, p. 109758, 2025.

[4] B. Zhai, G. J. Elder, and A. Godfrey, "Challenges and opportunities of deep learning for wearable-based objective sleep assessment," *npj Digital Medicine*, vol. 7, no. 1, p. 85, 2024.

[5] Y. Baek, B. Bae, H. Shin, C. Sonnadara, H. Cho, C.-Y. Lin, Y. Mu, C. Shen, S. Shah, G. Wang *et al.*, "Edge intelligence through in-sensor and near-sensor computing for the artificial intelligence of things," *npj Unconventional Computing*, vol. 2, no. 1, p. 25, 2025.

[6] H. A. Imran, Q. Riaz, K. Hamza, S. Muhammad, and B. Krüger, "From steps to sentiments: Cross-domain transfer learning for activity-based emotion detection in wearable iot systems," *IEEE Internet of Things Journal*, pp. 1–1, 2026.

[7] U. Khadam, P. Davidsson, and R. Spalazzese, "A systematic literature review on ai in iot systems: Tasks, applications, and deployment," *Internet of Things*, p. 101779, 2025.

[8] G.-Q. Zhang, L. Cui, R. Mueller, S. Tao, M. Kim, M. Rueschman, S. Mariani, D. Mobley, and S. Redline, "The national sleep research resource: towards a sleep data commons," *Journal of the American Medical Informatics Association*, vol. 25, no. 10, pp. 1351–1358, 2018.

[9] J. Palotti, R. Mall, M. Aupetit, M. Rueschman, M. Singh, A. Sathyanarayana, S. Taheri, and L. Fernandez-Luque, "Benchmark on a large cohort for sleep-wake classification with machine learning techniques," *NPJ digital medicine*, vol. 2, no. 1, p. 50, 2019.

[10] "Mesa sleep dataset," https://sleepdata.org/datasets/mesa, 2018, data made available by the National Sleep Research Resource (NSRR).

[11] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan *et al.*, "Searching for mobilenetv3," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 1314–1324.

Acknowledgements

- The Multi-Ethnic Study of Atherosclerosis (MESA) Sleep Ancillary study was funded by NIH-NHLBI Association of Sleep Disorders with Cardiovascular Health Across Ethnic Groups (RO1 HL098433). MESA is supported by NHLBI funded contracts HHSN268201500003I, N01-HC-95159, N01-HC-95160, N01-HC-95161, N01-HC-95162, N01-HC-95163, N01-HC-95164, N01-HC-95165, N01-HC-95166, N01-HC-95167, N01-HC-95168 and N01-HC-95169 from the National Heart, Lung, and Blood Institute, and by cooperative agreements UL1-TR-000040, UL1-TR-001079, and UL1-TR-001420 funded by NCATS. The National Sleep Research Resource was supported by the National Heart, Lung, and Blood Institute (R24 HL114473, 75N92019R002).
- AI-assisted tools were used to improve language and readability; the authors reviewed the content and take full responsibility.